

COPA amplifier tests summary

date: 18/06/2024
Place: PTB

Authors: RI - LP
version: V01

Two indential boards were built by CTU and delivered to PTB:
1) Board #1
2) Board #2

Tests on board #2:
a) Sweeps: -Gain, Phase, FFT, THD+N and SINAD

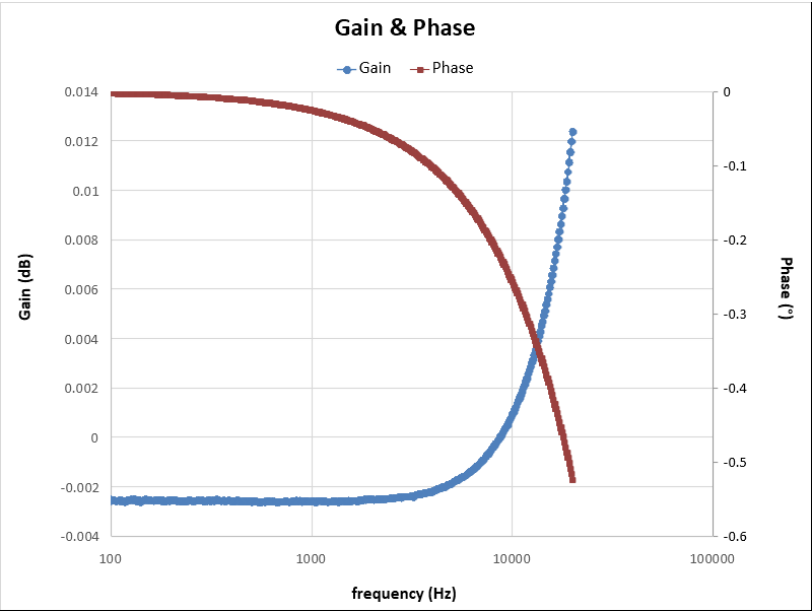


Figure XX. Gain and Phase Board_2 20 Hz to 20 kHz, BW DC-90 kHz.

b) THD:

Board_2	fo (Hz)	THD (dB)
	62.5	-143.52
	125	-138.58
	625	-127.85
	1000	-123.80
	2000	-117.42
	5000	-108.54
	10002	-98.97
	20006	-88.52
	20006	-88.33
	49986	-73.00
	49985	-73.09
	100026	-46.16
	100026	-45.01

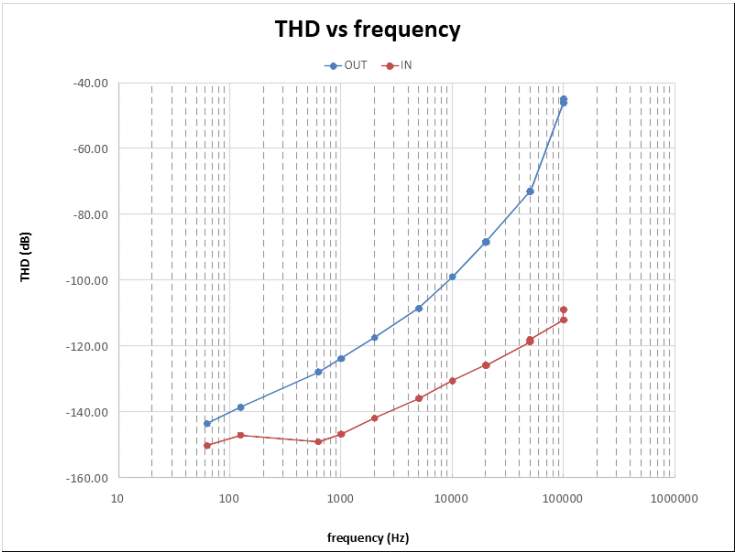


Figure XX. Total Harmonic Distortion Board_2

Tests on Board #1:

Board #1 was modified from its original version. Changes: i) the gain resistors R7 and R8 were removed, and the bootstrap circuit was disconnected and its input grounded.

Starting date: 13/06/2024

a) Total Harmonic Distortion

Board_1	fo (Hz)	THD (dB)
	62.5	-148.51
	125	-144.37
	625	-137.56
	1000	-134.14
	2000	-127.30
	5000	-115.49
	10000	-103.88
	20000	-92.43
	50000	-76.49
	100000	-45.14

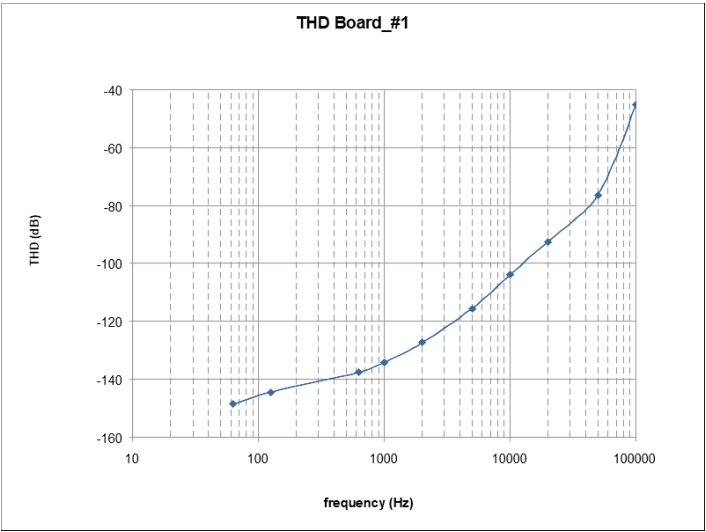


Figure XX. Total Harmonic Distortion Board_1

There was a improvement on the performance as it can be seen on the following figure were the THD are compared.

Total Harmonic distortion comparison Board_#1 and Board_#2

diff B2-B1	
fo (Hz)	(dB)
62.5	4.999
125	5.783
625	9.708
1000	10.343
2000	9.885
5000	6.947
10000	4.911
20000	3.913
50000	3.488
100000	-1.021

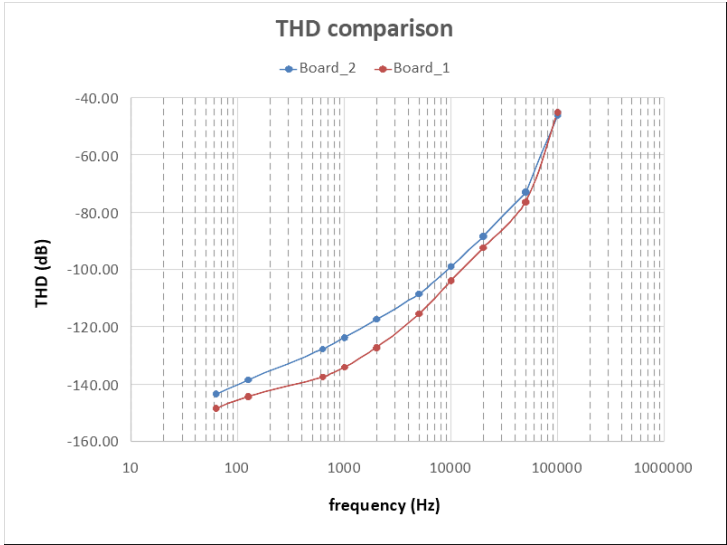


Figure XX. Total harmonic distortion comparison

b) Total Harmonic Distortion vs Input Amplitude. Decreasing the input amplitude, starting from the full scale (7.071 V rms) down to the value where the THD is less or equal to -120 dB
 At signal frequencies above 20 kHz it is not possible to get a THD < -120 dB
 for frequencies in the range of 62.5 Hz up to 2 kHz, not shown in the figure, the THD is lower than -120 dB.

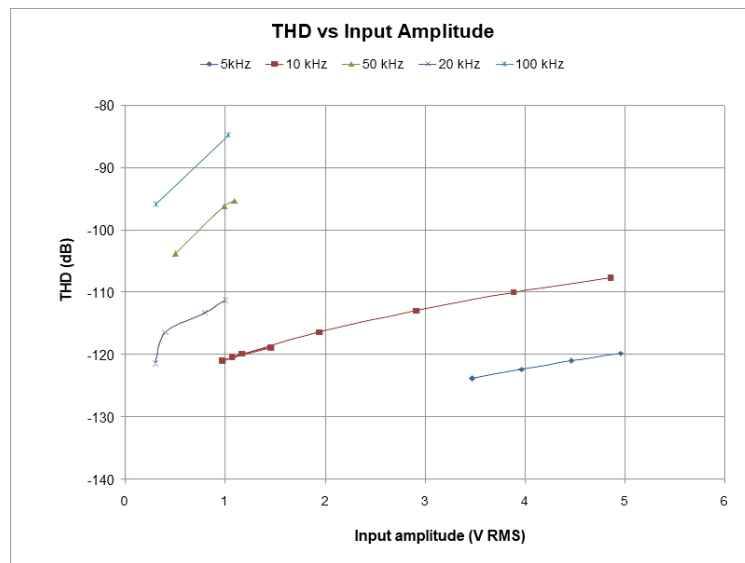


Figure XX. Total Harmonic Distortion versus input signal amplitude for frequencies in the range 5 kHz up to 100 kHz

c) Sweep Gain, Phase

Sweep from 20 Hz to 200 kHz, sampler BW DC - 1MHz, Sampling Rate: 2.496 MHz

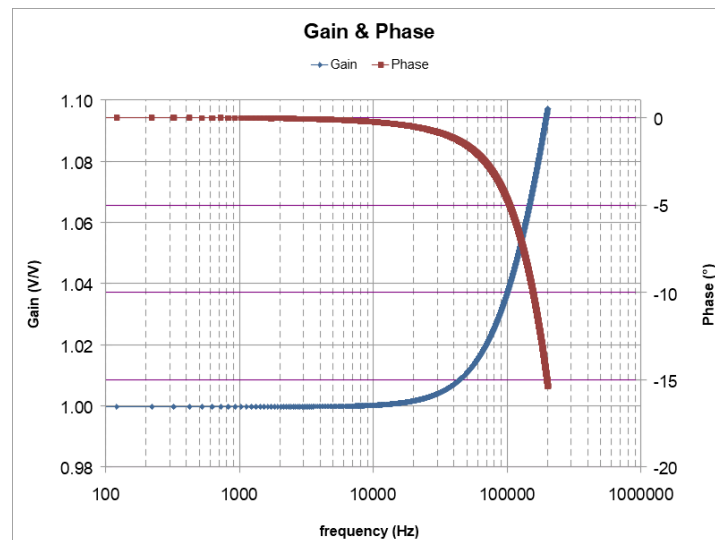


Figure XX. Board #1 Gain vs frequency.

d) Noise.

i) Voltage noise.

Setup: The COPA amplifier input was shorted and the output connected to a pre-amplifier (PTB PA3 Ultra Low Noise AC pre-amplifier) running both on batteries. The PTB PA3 internal batteries and the COPA powered by a +/-12 V battery pack. The PA3 output was connected to the AD CH2 sampler. The AP software was configured to sample the data at 192 kHz (BWDC 90 kHz), averaging 64 windows of 1.2 M data points with a rectangular window (no window function on the software).

AP nominal gain = x60, measured gain = x59.2

Figure XX shows the Amplitude Spectrum Density of the measured signal divided by the PA3 gain

factor.

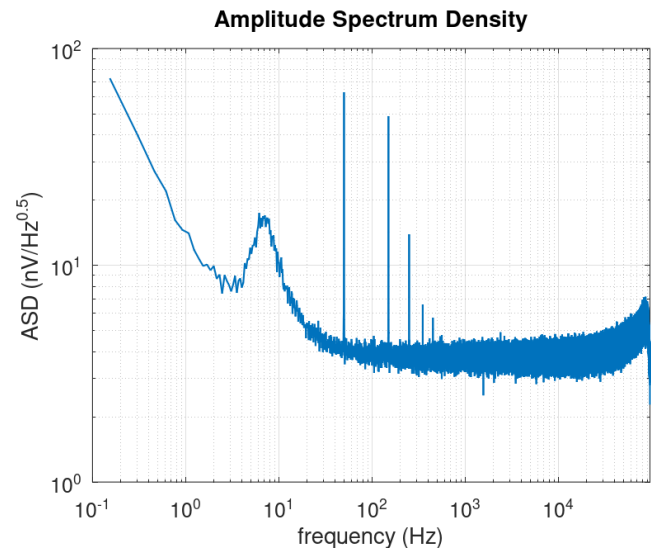
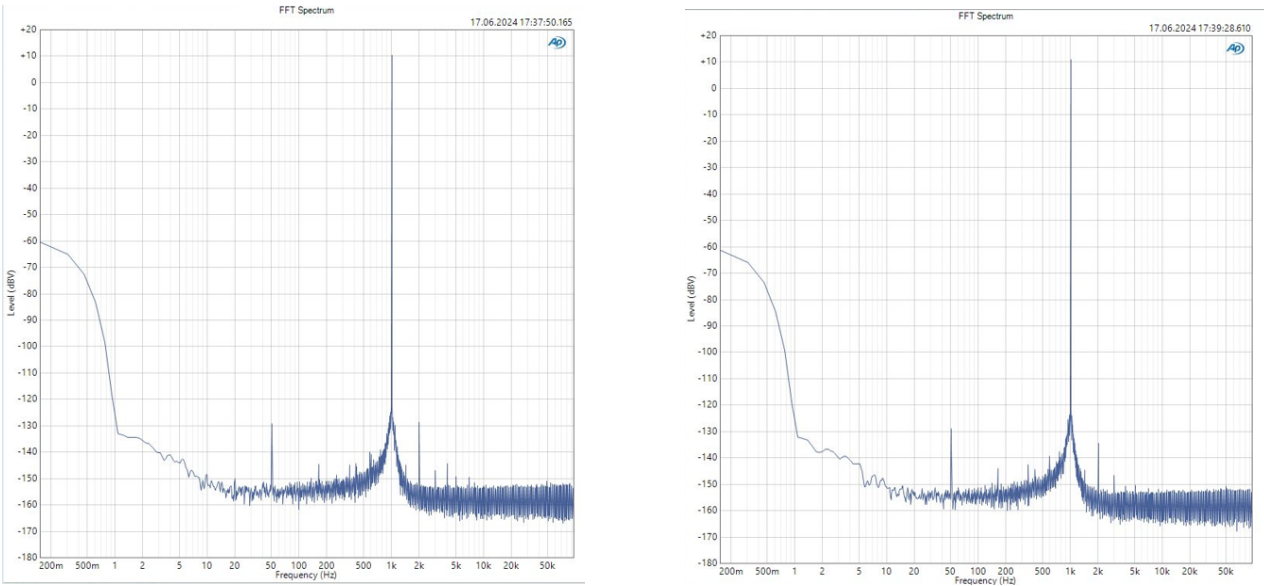


Figure XX. Board #1. Amplitude Spectrum Density over a 64 averaged (1.2M window length)

e) Spectrum at different loads values. In order to verify the load influence on the COPA amplifier spectrum, two spectras were acquried, with the AP sampler at 100 kΩ and 600 Ω.

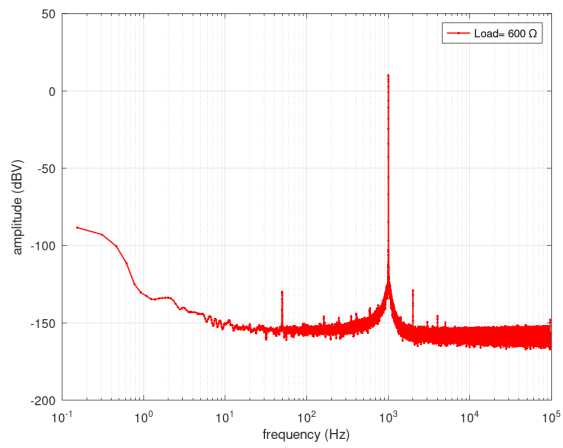


3.5 V RMS @ 1 kHz, 600 Ω.

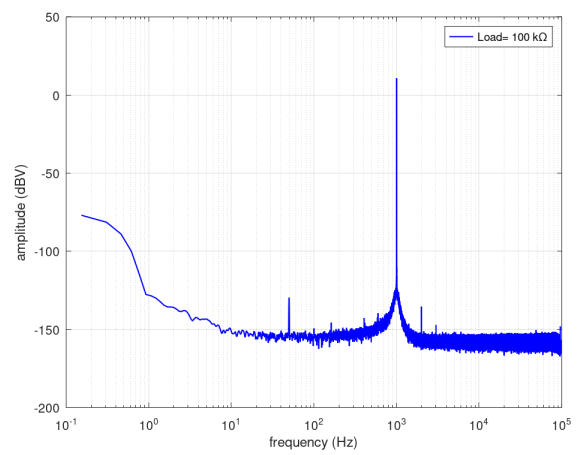
3.5 V RMS @ 1 kHz, 100 kΩ.

	freq (Hz)	600 OHM LOAD		100 KOHM LOAD		Delta (V)
		Amp (dBV)	Amp (V)	Amp (dBV)	Amp (V)	
fo	1.0002E+03	-0.82809	0.909066	-0.12816	0.985353	0.07629
2fo	2.0002E+03	-149.05	3.53E-08	-144.15	6.20E-08	2.67E-08
3fo	3.0003E+03	-159	1.12E-08	-156.63	1.47E-08	3.52E-09

Table for 1 V rms at 1 kHz



3.5 V RMS @ 1 kHz, 600 Ω.



3.5 V RMS @ 1 kHz, 100 kΩ.

	freq (Hz)	600 OHM LOAD		100 KOHM LOAD		Delta (V)
		Amp (dBV)	Amp (V)	Amp (dBV)	Amp (V)	
fo	1000.154	10.049	3.180168	10.748	3.446672	0.2665
2fo	2000.154	-128.985	3.55E-07	-135.39	1.70E-07	-1.85E-07
3fo	3000.308	-149.577	3.32E-08	-147.197	4.37E-08	1.05E-08
4fo	4000.462	-145.531	5.29E-08	-154.121	1.97E-08	-3.32E-08

Table for 3.5 V rms at 1 kHz

SIMULATION Results
Gain and Phase LTSPICE

